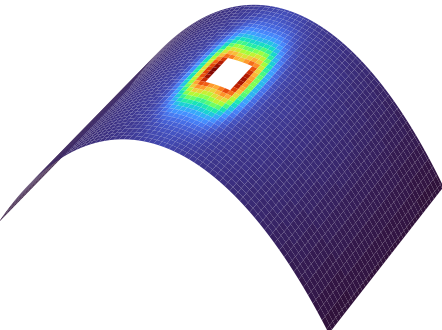


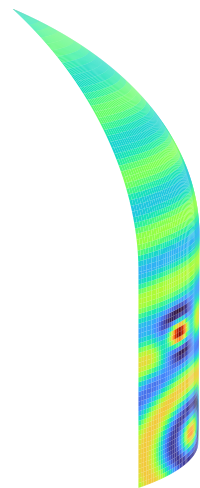
A structure-agnostic SHM decision framework across five reusable CFRP spacecraft structures and four load modes

Perforated interstage



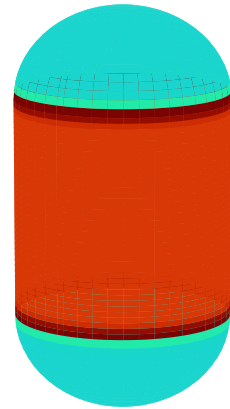
load: in-plane / membrane
sensing: surface-stress GNN

H3 fairing



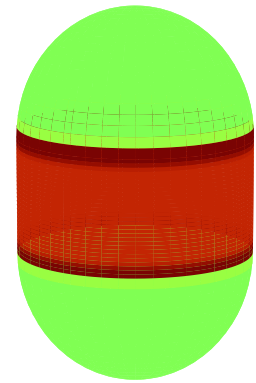
load: transverse press. / bending
sensing: guided-wave GNN

SRB-3 motor case



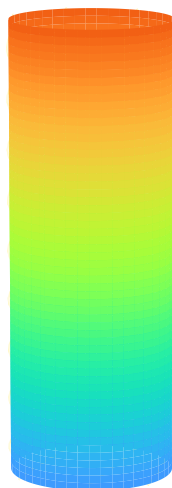
load: internal-pressure burst
sensing: acoustic emission

CFRP COPV tank



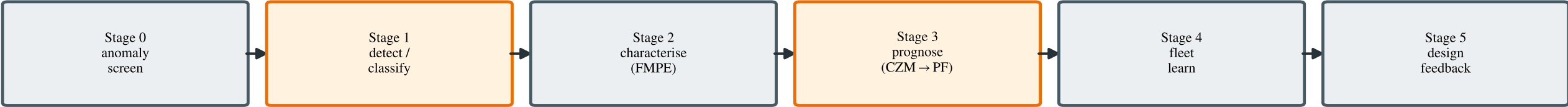
load: internal pressure
sensing: shared $|z|$ screen

CFRP drive shaft



load: TORSION (shear)
sensing: shared $|z|$ screen

Shared structure-agnostic decision core — only detection (Stage 1) & prognosis physics (Stage 3) are structure-specific (pluggable)



structure-agnostic

structure-specific

structure-agnostic

structure-specific

structure-agnostic

structure-agnostic

detect → classify → characterise → prognose → clear (OK/REPAIR/RETIRE) → fleet-learn → design | 1,499 real-FEM states, one margin surrogate $R^2=0.94$